

BikeShare Data Warehouse & Business Intelligence Solution

Contents

1.Data Source	3
1.1 Data Source for Assignment 2	3
2. SSAS Cube Implementation	4
2.1 Creating the SSAS Project.....	4
2.2 Creating the Data Source	5
2.3 Creating the Data Source View	5
2.4 Creating Dimensions	6
2.5 Creating the Cube	6
3. Demonstration of OLAP Operations	8
3.1 Roll-up.....	8
3.2 Drill-down.....	9
3.3 Slice	10
3.4 Dice.....	11
3.5 Pivot.....	11
4. Power BI Reports.....	12
4.1 Report 1 — Matrix Visual	12
4.2 Report 2 — Cascading Slicers	13
4.3 Report 3 — Drill-down Report.....	14
4.4 Report 4 — Drill-through Report	15

1.Data Source

1.1 Data Source for Assignment 2

The data source used for Assignment 2 is the BikeDW data warehouse that was designed and implemented in Assignment 1. This data warehouse was built using SQL Server and follows a star schema dimensional model.

It serves as the analytical data store for the SSAS cube and Power BI reports developed in this assignment.

The data warehouse consists of the following tables:

Fact Table:

- Fact_Trips: Contains over 108,000 bike trip records covering all 12 months of 2023. Each record represents a single bike trip and includes measures such as trip_duration (in minutes) and txn_process_time_hours (transaction processing time). The table also contains foreign keys referencing the dimension tables.

Dimension Tables:

- Dim_Date: Contains 365 date records for the year 2023, with attributes including date_key, full_date, year, month, and day. This dimension supports time-based analysis and hierarchical drill-down from year to month to day.

- Dim_Station: Contains bike station information including station_key, station_name, latitude, longitude, and SCD Type 2 columns (start_date, end_date, current_flag). This dimension implements Slowly Changing Dimension Type 2 to track historical changes in station attributes.

- Dim_User: Contains user type information with two categories — member and casual. This dimension enables customer segmentation analysis.

The star schema design connects Fact_Trips to all three dimension tables through surrogate keys, enabling efficient analytical querying and multidimensional analysis.

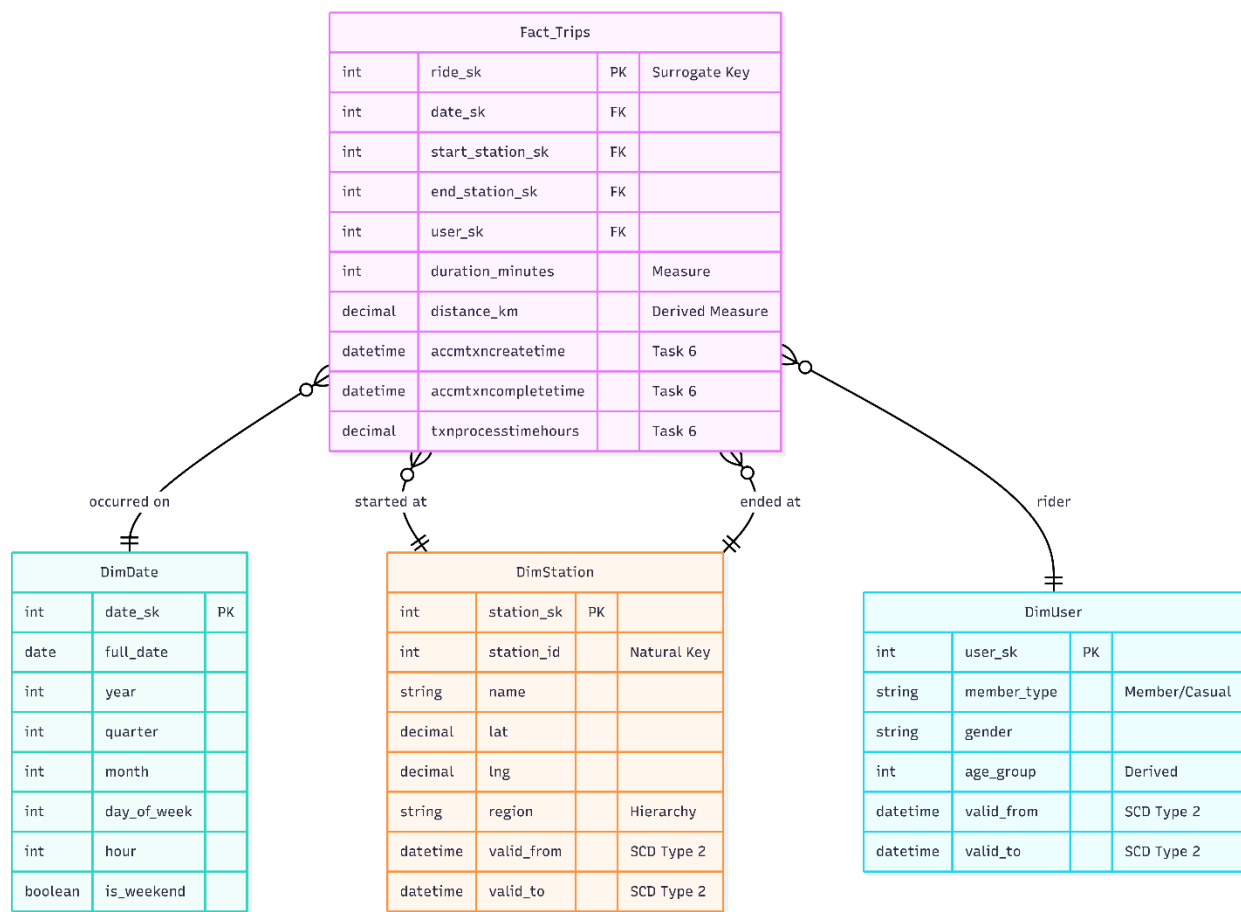


Figure 1- Star Schema

2. SSAS Cube Implementation

This section describes the steps followed to design and implement an SSAS (SQL Server Analysis Services) Multidimensional cube using the BikeDW data warehouse as the data source.

2.1 Creating the SSAS Project

A new Analysis Services Multidimensional and Data Mining Project was created in Visual Studio using the Microsoft Analysis Services Projects 2022 extension. The project was named "MultidimensionalProject-Bike".

The project structure in Visual Studio contains the following components:

- Data Sources: Connection to the BikeDW SQL Server database
- Data Source Views: Visual representation of the tables and relationships
- Cubes: The multidimensional cube definition

- Dimensions: Individual dimension definitions

2.2 Creating the Data Source

A new data source was created by connecting to the BikeDW SQL Server database on the NETHMAA21\MSSQLSERVER2 server instance. Windows Authentication (SSPI) was used for the connection. The data source was named "Bike DW".

2.3 Creating the Data Source View

A Data Source View (DSV) was created to define the logical structure of the data used by the cube. The following four tables were added from the BikeDW database:

- Fact_Trips (fact table)
- Dim_Date (date dimension)
- Dim_Station (station dimension)
- Dim_User (user dimension)

Relationships were manually defined between the fact table and each dimension table:

- Fact_Trips.date_key → Dim_Date.date_key
- Fact_Trips.user_key → Dim_User.user_key
- Fact_Trips.start_station_key → Dim_Station.station_key
- Fact_Trips.end_station_key → Dim_Station.station_key

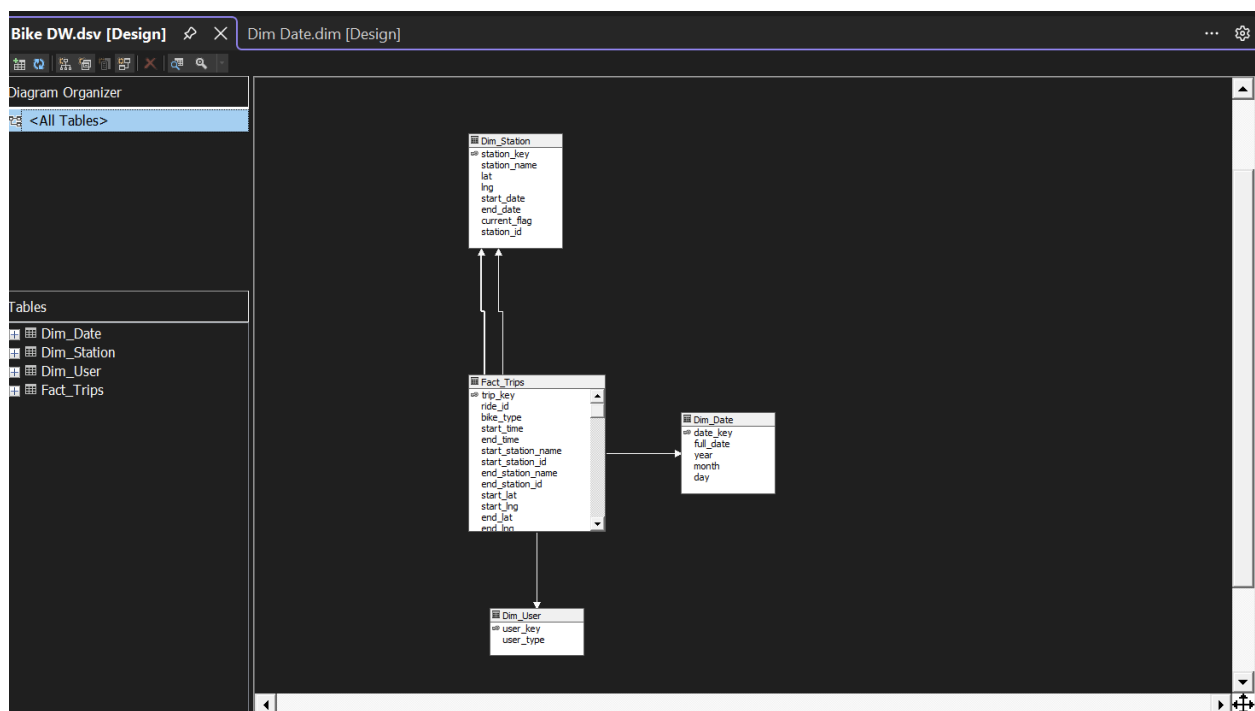


Figure 2 - Data Source View showing tables and relationships

2.4 Creating Dimensions

Three dimensions were created from the Data Source View:

Dim_Date Dimension:

The date dimension was created with the following attributes:

- Date Key (surrogate key)
- Year
- Month
- Day
- Full Date

A user-defined hierarchy named "Date Hierarchy" was created with three levels: Year → Month → Day. This hierarchy enables hierarchical analysis and drill-down operations from yearly to monthly to daily level. Composite keys were used for Month (year+month) and Day (year+month+day) to ensure uniqueness across years.

Dim_Station Dimension:

The station dimension was created with the following attributes:

- Station Key (surrogate key)
- Station Name

This dimension represents the bike stations where trips start and end. Two role-playing dimensions were created from Dim_Station — Start Station and End Station — to support analysis of both trip origins and destinations.

Dim_User Dimension:

The user dimension was created with the following attributes:

- User Key (surrogate key)
- User Type (member/casual)

This dimension enables customer segmentation analysis between registered members and casual users.

2.5 Creating the Cube

The cube was created using the existing tables from the Data Source View. The following components were configured:

Measures (from Fact_Trips):

- Trip Duration: Sum of trip duration in minutes
- Txn Process Time Hours: Sum of transaction processing time in hours
- Fact Trips Count: Count of total trips

Dimensions connected to the cube:

- Dim User
- Dim Date (with Date Hierarchy)
- Start Station (role-playing dimension)
- End Station (role-playing dimension)

2.6 Deployment and Processing

Due to a version compatibility issue between the Analysis Services Projects extension (v4.0.0) and the SQL Server Analysis Services instance (v17.0), the cube could not be deployed directly from Visual Studio. Instead, the cube was deployed using an XMLA script executed through SQL Server Management Studio (SSMS).

The XMLA script was generated from the project's .asdatabase file and wrapped in a Create command. The cube was successfully deployed to the NETHMAA21\MSSQLSERVER2 Analysis Services instance.

After deployment, the cube was processed using a Process Full command to load data from the BikeDW database into the SSAS cube.

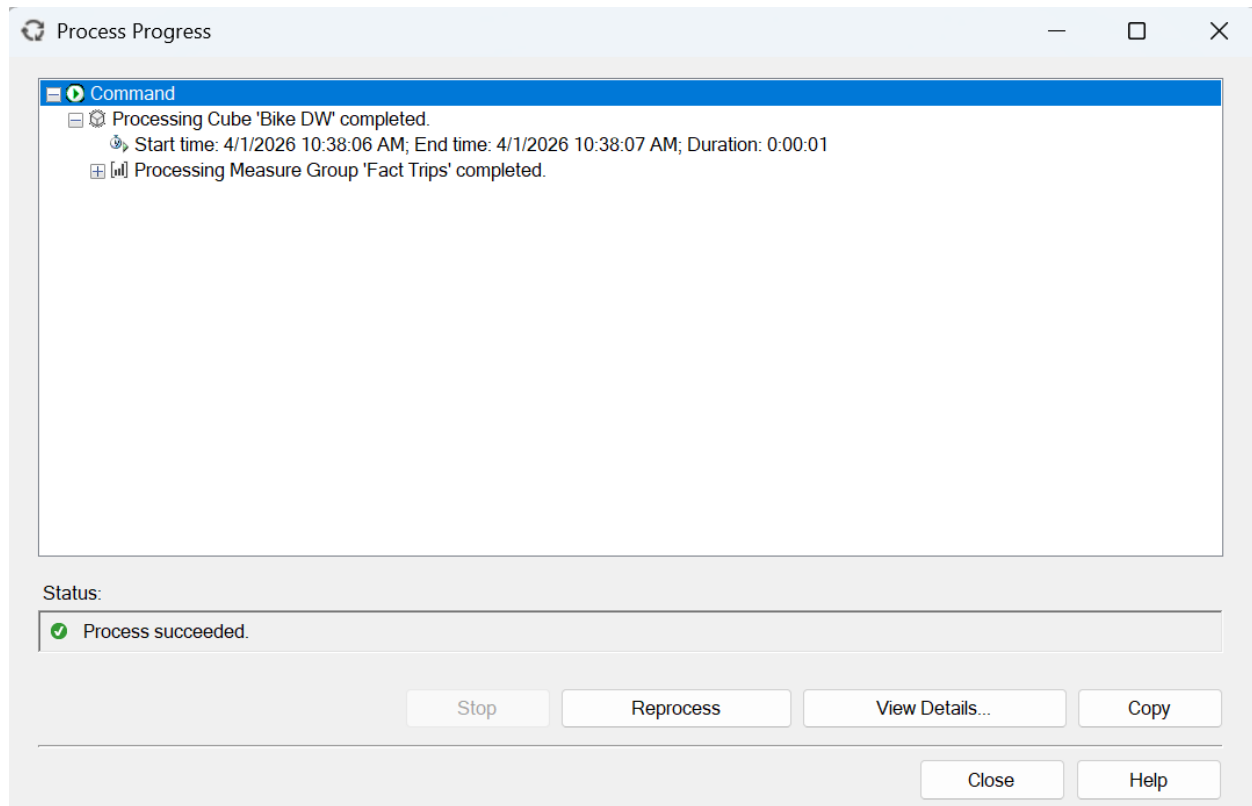


Figure 3 - SSMS Process progress

3. Demonstration of OLAP Operations

OLAP operations were demonstrated by connecting an Excel workbook to the SSAS cube using the Data tab → Get Data → From Analysis Services feature. The server NETHMAA21\MSSQLSERVER2 was used to connect to the MultidimensionalProject-Bike database and Bike DW cube.

A PivotTable was created in Excel to perform the following OLAP operations:

3.1 Roll-up

Roll-up is an OLAP operation that aggregates data by moving up the hierarchy from a lower level of detail to a higher level. In this demonstration, trip duration data was rolled up from the monthly level to the yearly level using the Date Hierarchy. The PivotTable shows the total sum of trip duration for the entire year 2023, aggregating all monthly values into a single total.

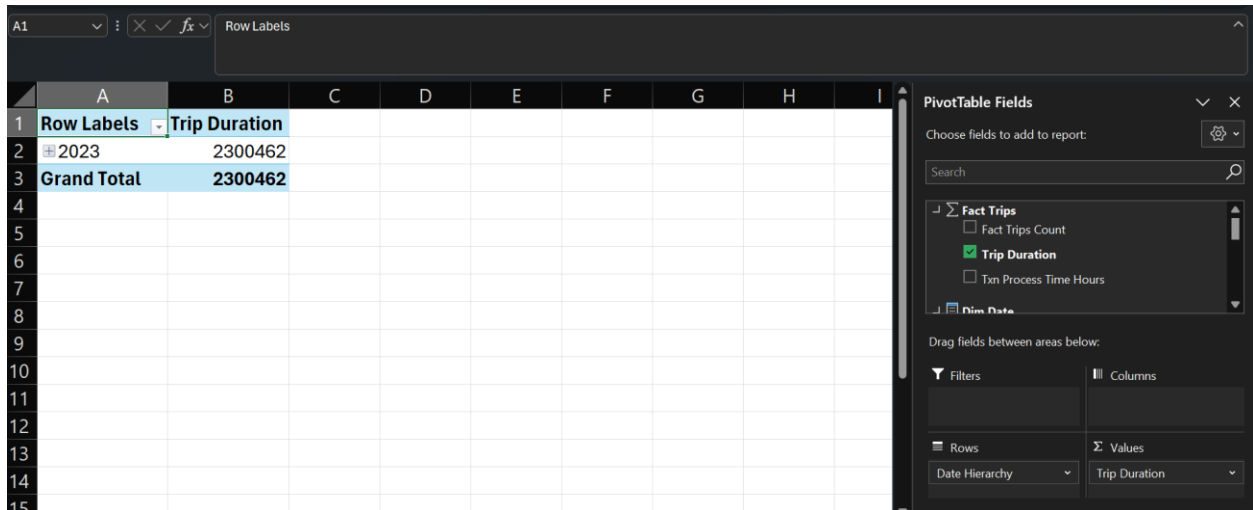


Figure 4 - Roll up

3.2 Drill-down

Drill-down is the opposite of roll-up — it navigates from a higher level of aggregation to a lower level of detail. Starting from the Year level showing 2023 totals, the hierarchy was expanded to show monthly trip duration values (January through December). Further expansion revealed daily trip duration values within each month, demonstrating the full Year → Month → Day drill-down path through the Date Hierarchy.

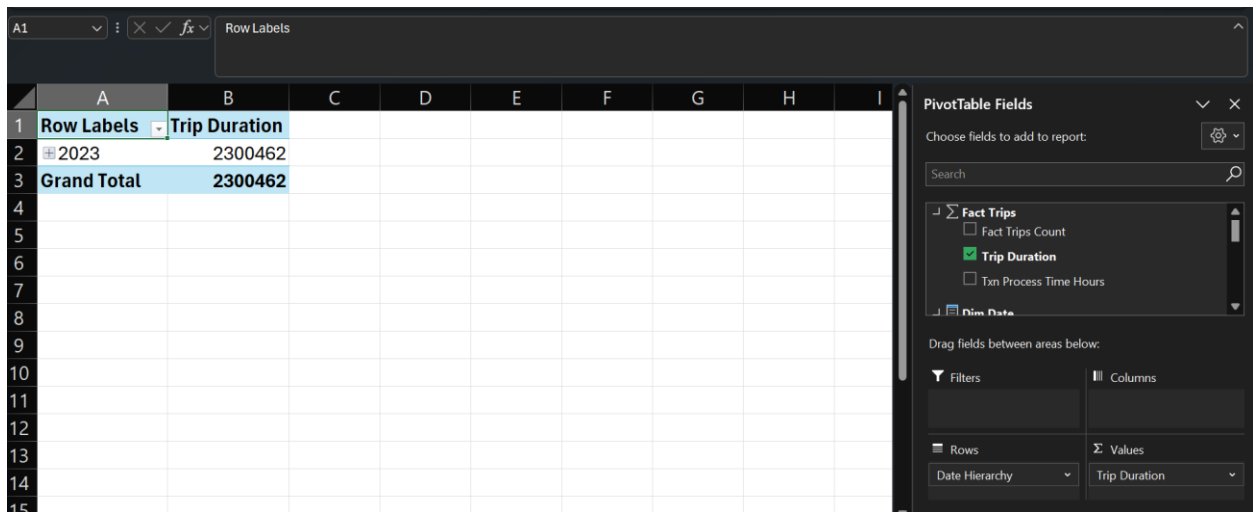


Figure 5 - Drill down 1

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
	Row Labels	Trip Duration															
	2023																
	1																
	1																
	2023-01-01																
	1	11140															
	2023-01-02																
	1	35630															
	2023-01-03																
	1	6071															
	2023-01-04																
	1	4387															
	2023-01-05																
	1	8213															
	2023-01-06																
	1	7652															
	2023-01-07																
	1	9994															
	2023-01-08																
	1	5186															
	2023-01-09																
	1	3575															
	2023-01-10																
	1	7476															
	2023-01-11																
	1	5301															
	2023-01-12																
	1	1013															
	2023-01-13																
	1	11633															
	2023-01-14																

Figure 6 - Drill down 2

3.3 Slice

Slice is an OLAP operation that selects a single value from one dimension to filter the entire dataset. In this demonstration, the User Type dimension was added to the PivotTable filter area.

The filter was set to "member" only, which sliced the data to show trip duration values exclusively for registered member users, removing all casual user data from the view.

C4																	
1	User Type	member															
2																	
3	Row Labels	Trip Duration															
4	2023																
5	1																
6	1	997987															
7	Grand Total	997987															
8																	
9																	
10																	
11																	
12																	
13																	
14																	
15																	
16																	
17																	
18																	
19																	
20																	

Figure 7 - Slice

3.4 Dice

Dice is similar to slice but applies filters on multiple dimensions simultaneously to create a subcube. In this demonstration, two filters were applied at the same time — User Type was filtered to "member" AND Month was filtered to specific months (e.g., January and February). This produced a view showing only member user trips during the selected months, demonstrating multi-dimensional filtering.

Row Labels	Trip Duration
2023	
1	997987
Grand Total	997987

Figure 8 - Dice

3.5 Pivot

Pivot (also called rotate) changes the dimensional orientation of the data by swapping rows and columns. Initially, the PivotTable displayed station names in rows and trip duration as values. The User Type dimension was then moved from the Rows area to the Columns area, rotating the data perspective to show trip duration broken down by user type (casual vs member) across different stations.

Row Labels	Trip Duration
2023	
1	
casual	21962
member	67875
10th & E St NW	
casual	2670
member	1067
10th & G St NW	
casual	1814
member	2978
10th & H St NE	
casual	934
member	2387
10th & K St NW	
casual	4511
member	3039
10th & Monroe St NE	

Figure 9 - Pivot (before)

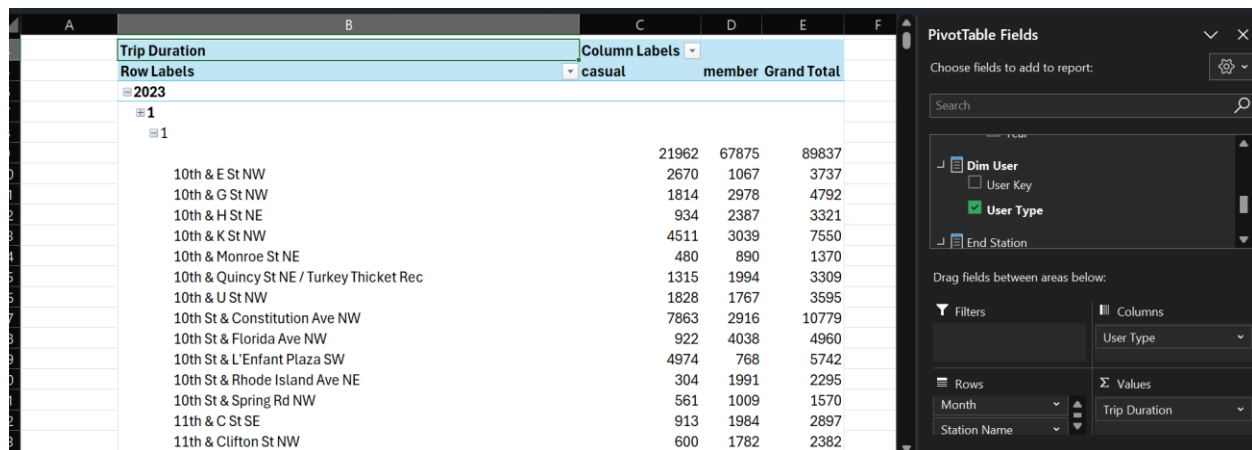


Figure 10 - Pivot (after)

4. Power BI Reports

Four reports were developed in Power BI Desktop and published to Power BI Service using the SLIIT student Microsoft account. The reports were created by connecting Power BI directly to the BikeDW SQL Server database and importing Fact_Trips, Dim_Date, Dim_Station, and Dim_User tables.

Data Preparation:

After importing the tables, relationships were verified in the Model view. The relationship between Fact_Trips and Dim_Station was manually created using start_station_key → station_key (active) and end_station_key → station_key (inactive).

4.1 Report 1 — Matrix Visual

A Matrix visual was created to display detailed tabular data with row and column groupings. The matrix shows the sum of trip duration broken down by station name (rows) and user type — casual and member (columns). Grand totals are shown for both rows and columns, providing a comprehensive view of trip patterns across all stations.

The matrix clearly shows that member users generally have higher trip duration totals compared to casual users across most stations, with the overall total reaching 2,300,462 minutes across all stations.

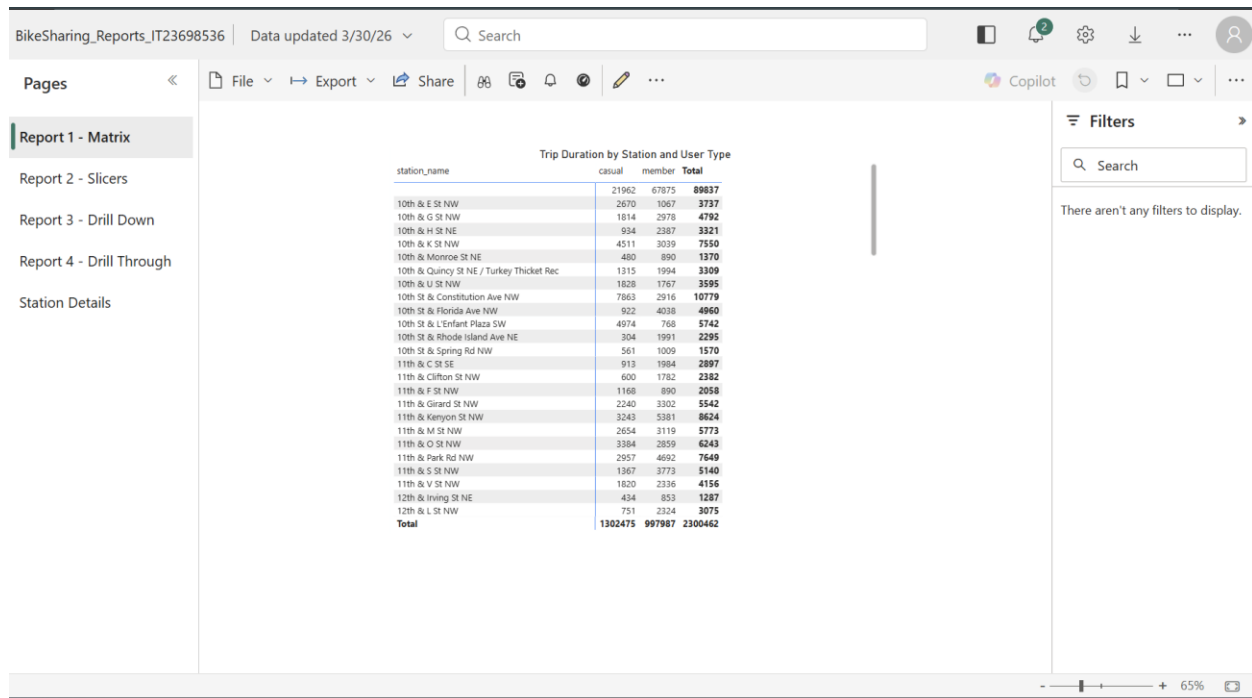


Figure 11 - Report 1

4.2 Report 2 — Cascading Slicers

This report demonstrates cascading filters using two slicers. A Year slicer and a Month slicer were added to the report canvas. The cascading filter was implemented using the Edit Interactions feature — selecting a year in the first slicer dynamically filters the available months in the second slicer, ensuring only relevant months are shown.

Two visualizations were added to the report:

1. Clustered Bar Chart: Displays sum of trip duration by station name, showing the top stations by total trip duration.
2. Pie Chart: Shows the distribution of trip duration between casual and member user types.

Both charts respond dynamically to slicer selections, updating in real time when year or month filters are applied.

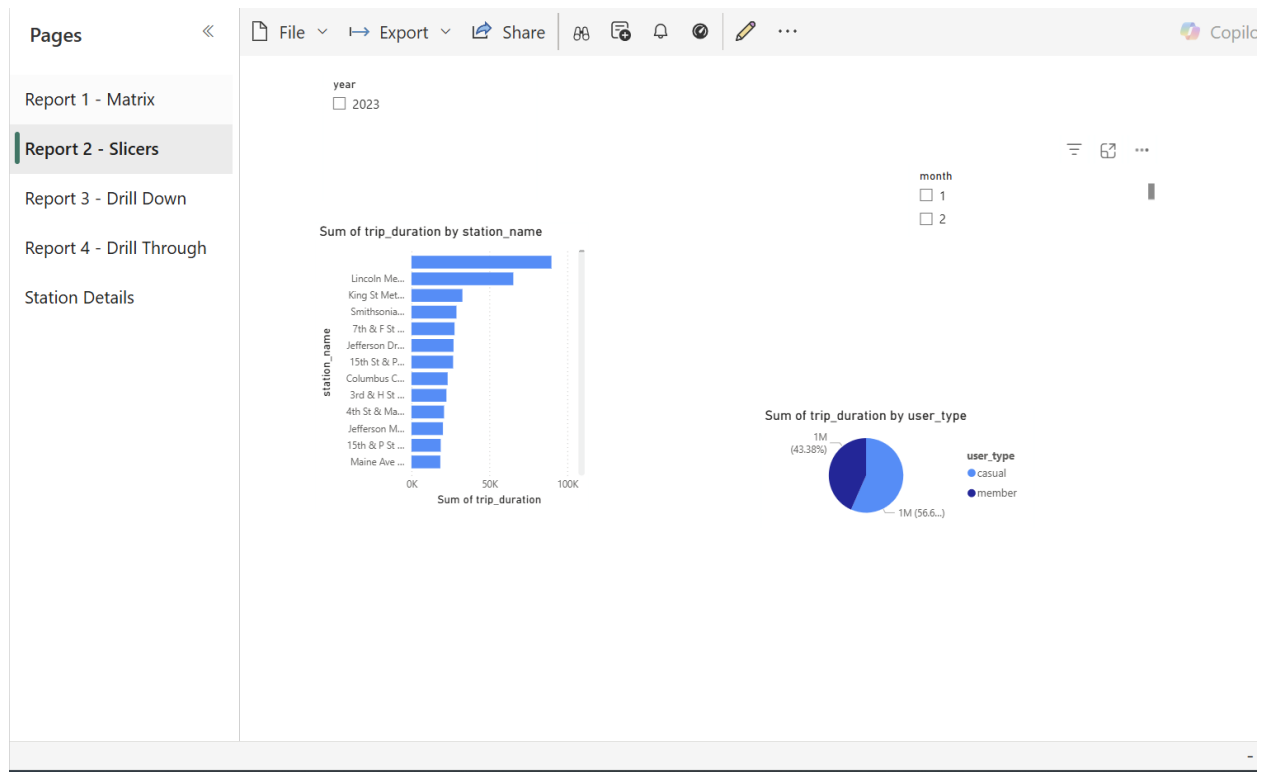


Figure 12 - Report 2

4.3 Report 3 — Drill-down Report

A Clustered Column Chart was created with a date hierarchy for drill-down analysis. The Year, Month, and Day fields from Dim_Date were added to the X-axis, creating a hierarchical axis that allows users to explore trip duration data at different time granularities.

Users can drill down by:

1. Clicking the drill-down arrow (↓) to enable drill mode
2. Clicking on a bar to drill from Year → Month → Day level
3. Using the drill-up arrow (↑) to return to a higher level

The report title "Trip Trends: Year → Month → Day" clearly communicates the drill-down path to users.

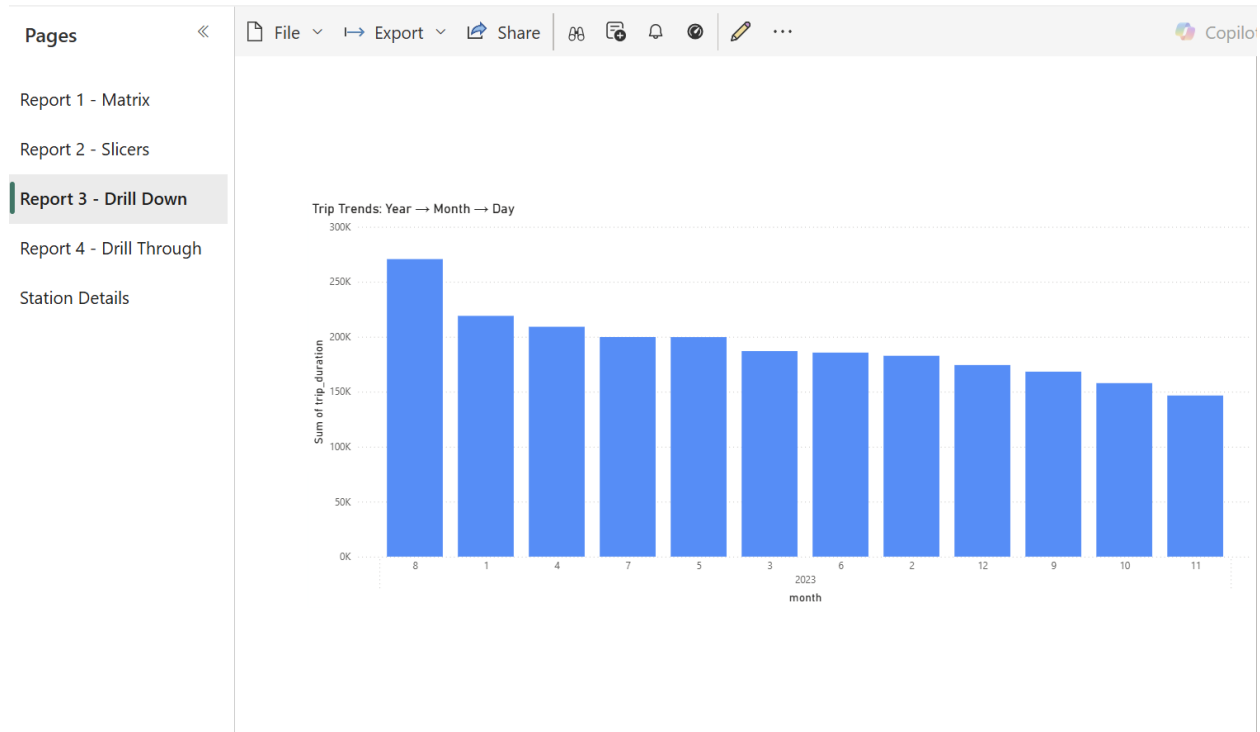


Figure 13 - Report 3

4.4 Report 4 — Drill-through Report

This report consists of two pages implementing drill-through navigation:

Page 1 (Report 4 - Drill Through): A summary page containing a Clustered Bar Chart showing sum of trip duration by user type (casual and member). This serves as the starting point for drill-through navigation.

Page 2 (Station Details): A detail page configured as the drill-through destination. It contains a Table visual showing station name, trip duration, and user type for detailed analysis. The `user_type` field was added to the Drill Through field well, making this page a drill-through target.

To use the drill-through: Right-click on a bar in the summary page → select Drill Through → Station Details. The Station Details page opens filtered by the selected user type, showing detailed trip information for that user category.

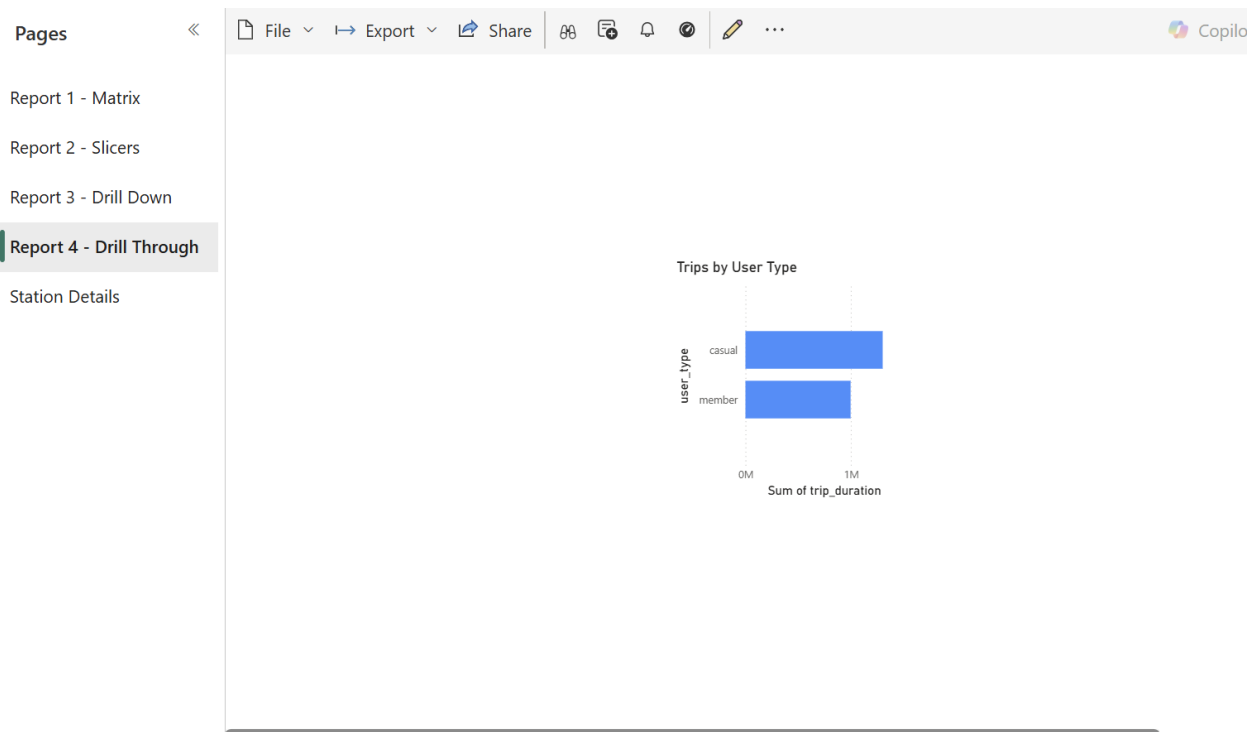


Figure 14 – Summary page

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Report 1 - Matrix

Report 2 - Slicers

Report 3 - Drill Down

Report 4 - Drill Through

Station Details

Station Details Selected

Station Detail View

station_name	Sum of trip_duration	user_type
10th & E St NW	21962	casual
10th & G St NW	2670	casual
10th & H St NE	1814	casual
10th & K St NW	934	casual
10th & Monroe St NE	4511	casual
10th & Quincy St NE / Turkey Thicket Rec	480	casual
10th & U St NW	1315	casual
10th St & Constitution Ave NW	1828	casual
10th St & Florida Ave NW	7863	casual
10th St & L'Enfant Plaza SW	922	casual
10th St & Rhode Island Ave NE	4974	casual
10th St & Spring Rd NW	304	casual
11th & C St SE	561	casual
11th & Clifton St NW	913	casual
11th & F St NW	600	casual
11th & Girard St NW	1168	casual
11th & Kenyon St NW	2240	casual
11th & M St NW	3243	casual
Total	1302475	

Figure 15 - Station details page